- The sequence number encoded as a 5-byte big-endian integer.

The resulting array is then combined with the Common IV using bitwise XOR, using the following method:

```
private static byte[] xor(byte[] xs, byte[] ys) {
    for (int i = 0; i < xs.length; ++i)
        xs[i] ^= ys[i];
    return xs;
}
```

Add the `xor()` method and the `nonce()` method from listing 13.11 to the `Oscore` class.

> **NOTE** Although the generated nonce looks random due to being XORed with the Common IV, it is in fact a deterministic counter that changes predictably as the sequence number increases. The encoding is designed to reduce the risk of accidental nonce reuse.

Listing 13.11 Deriving the per message nonce

```
private static byte[] nonce(int ivLength, long
sequenceNumber,
                            byte[] id, byte[] commonIv) {
    if (sequenceNumber > (1L << 40))    #A
        throw new IllegalArgumentException(    #A
            "Sequence number too large");   #A
    int idLen = ivLength - 6;    #B
    if (id.length > idLen)    #B
        throw new IllegalArgumentException("ID is too
large");    #B
```